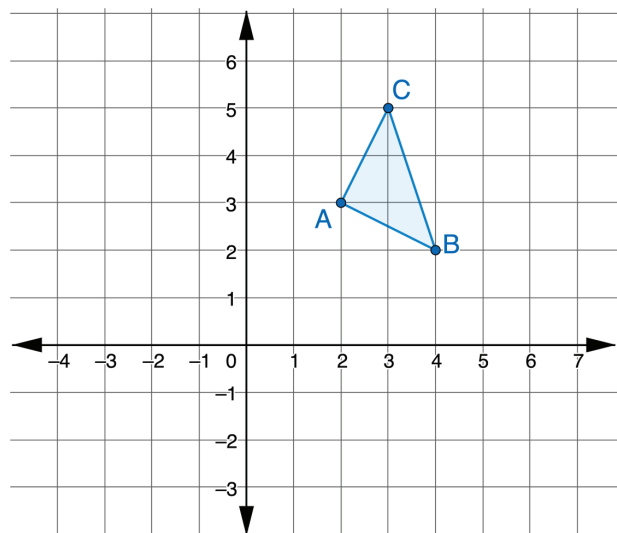




6.H3.3 Wrap-Up: Cool Down

1. $\triangle ABC$ is shown.



A sequence of three transformations is shown.

- $(x, y) \rightarrow (x - 4, y - 1)$
- $(x, y) \rightarrow (y, x)$
- $(x, y) \rightarrow (1.5x, 1.5y)$

- Draw the result of each transformation.
- Complete the table by noting if the preimage and image are congruent or similar.

Transformation	Relationship of Preimage to Image
$(x, y) \rightarrow (x - 4, y - 1)$	
$(x, y) \rightarrow (y, x)$	
$(x, y) \rightarrow (1.5x, 1.5y)$	

Unit 6, Lesson 10: Similar Figures



Benchmark

MA.912.GR.1.6 Solve mathematical and real-world problems involving congruence or similarity in two-dimensional figures.



Learning Target

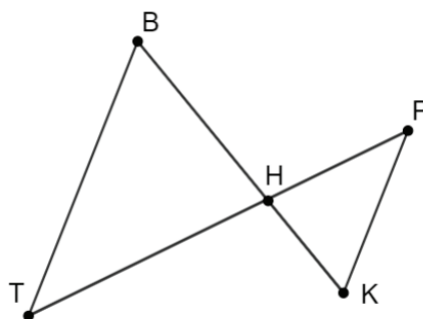
- ☐ I can solve mathematical problems using corresponding angles and sides of similar figures.



6.10.1 Warm-Up: Bell Work

1. Complete the proof.

Given: $\overline{BT} \parallel \overline{PK}$
 Prove: $\triangle BHT \sim \triangle KHP$



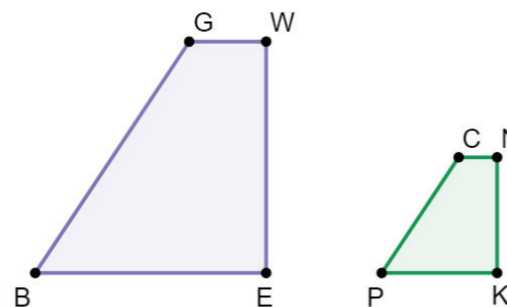
Statement	Reason
1. $\overline{BT} \parallel \overline{PK}$	1.
2.	2.
3.	3.
4. $\triangle BHT \sim \triangle KHP$	4.



6.10.2 Collaboration: Similar Figures

With your partner, answer the following questions.

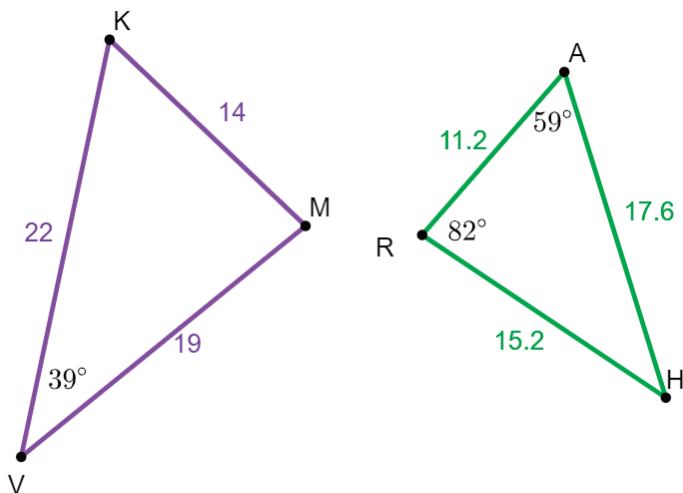
1. $GWEB$ and $CNKP$ are shown.



- If $GWEB \sim CNKP$, what is true about the corresponding angles?
- If $GWEB \sim CNKP$, what is true about the corresponding sides?
- Fill in the table given that $GWEB \sim CNKP$.

Corresponding Angles	Corresponding Sides
$\angle GBE \cong \angle$	$\frac{PK}{EB} = \frac{KN}{EW}$
$\angle PKN \cong \angle$	$\frac{CN}{EB} = \frac{PC}{EW}$
$\angle PCN \cong \angle$	$\frac{GB}{EB} = \frac{WG}{EW}$
$\angle GWE \cong \angle$	$\frac{PC}{EB} = \frac{PC}{EW}$

2. $\triangle VKM$ and $\triangle HRA$ are shown. $KM = 14$, $MV = 19$, $VK = 22$, $RA = 11.2$, $AH = 17.6$, $HR = 15.2$, $m\angle KVM = 39^\circ$, $m\angle HRA = 82^\circ$, and $m\angle RAH = 59^\circ$.



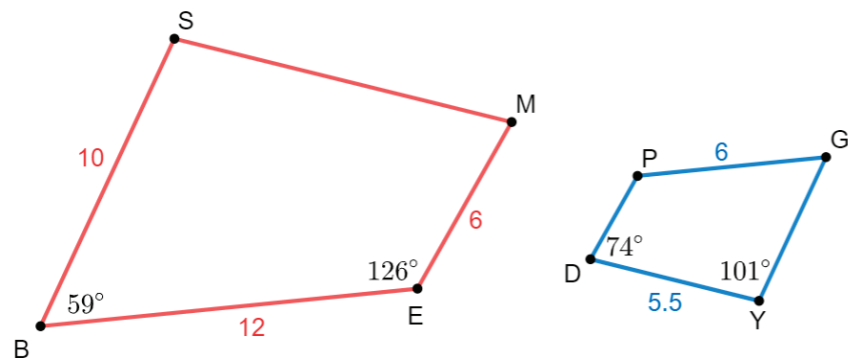
- Using the given information, what can you conclude about $\triangle VKM$ and $\triangle HRA$?
- What similarity theorem or postulate could be used to prove the triangles are similar?
- Write a similarity statement for $\triangle VKM$ and $\triangle HRA$.

- d. Complete the table.

Angle Measures	Side Proportions
$m\angle VKM =$	$\frac{MK}{RA} =$
$m\angle KVM =$	$\frac{MV}{RH} =$
$m\angle RHA =$	$\frac{KV}{AH} =$

- What scale factor are the sides of $\triangle VKM$ multiplied by to find the corresponding sides of $\triangle HRA$?
- If $MK = 21$, what should the length of $\overline{RA}$ be so that the triangles are similar?

3. $SMEB$ and $YDPG$ are shown. $SB = 10$, $BE = 12$, $EM = 6$, $DY = 5.5$, $PG = 6$, $m\angle SBE = 59^\circ$, $m\angle BEM = 126^\circ$, $m\angle PDY = 74^\circ$, and $m\angle DYG = 101^\circ$.



- a. If $SMEB \sim YDPG$, determine the measure of each.

$$m\angle BSM = \underline{\hspace{2cm}} \qquad m\angle SME = \underline{\hspace{2cm}}$$

$$m\angle DPG = \underline{\hspace{2cm}} \qquad m\angle PGY = \underline{\hspace{2cm}}$$

$$SM = \underline{\hspace{2cm}} \qquad PD = \underline{\hspace{2cm}}$$

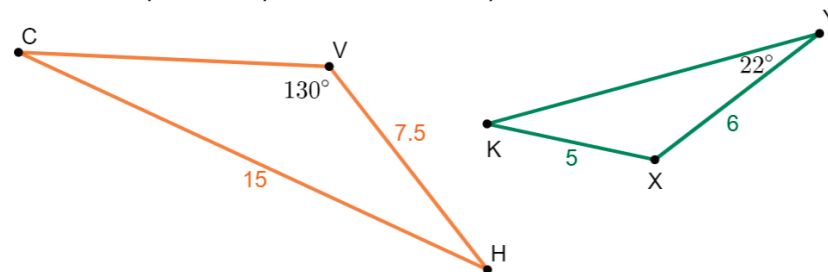
$$GY = \underline{\hspace{2cm}}$$

- b. Explain how the similarity relationship, $SMEB \sim YDPG$, helped determine which sides of the quadrilaterals are proportional.



6.10.3 Your Turn!

1. $\triangle CVH$ and $\triangle YXK$ are shown where $VH = 7.5$, $CH = 15$, $KX = 5$, $XY = 6$, $m\angle CVH = 130^\circ$, and $m\angle XYK = 22^\circ$.



- a. Given that $\triangle CVH \sim \triangle YXK$, determine the measure of each angle.

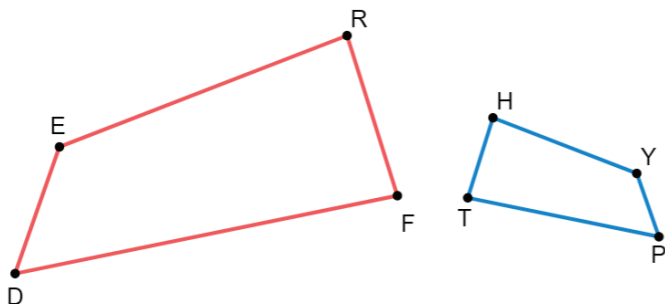
$$m\angle VCH = \underline{\hspace{2cm}} \qquad m\angle VHC = \underline{\hspace{2cm}}$$

$$m\angle YKX = \underline{\hspace{2cm}} \qquad m\angle YXK = \underline{\hspace{2cm}}$$

- b. Given that $\triangle CVH \sim \triangle YXK$, determine the measure of each side.

$$CV = \underline{\hspace{2cm}} \qquad KY = \underline{\hspace{2cm}}$$

2. $RFDE$ and $HTPY$ are shown, where $RE = 6.4$, $ED = 2.8$, $HY = 3.2$, $HT = 1.7$, $PT = 4$, $m\angle ERF = 86^\circ$, $m\angle EDF = 59^\circ$, and $m\angle HYP = 131^\circ$.



Assume: $RFDE \sim HTPY$

- a. Find the measure of $\angle DER$. Show your work or explain your thinking.

- b. Determine each measure.

$$m\angle HTP = \underline{\hspace{2cm}} \quad m\angle DFR = \underline{\hspace{2cm}}$$

$$m\angle THY = \underline{\hspace{2cm}} \quad m\angle YPT = \underline{\hspace{2cm}}$$

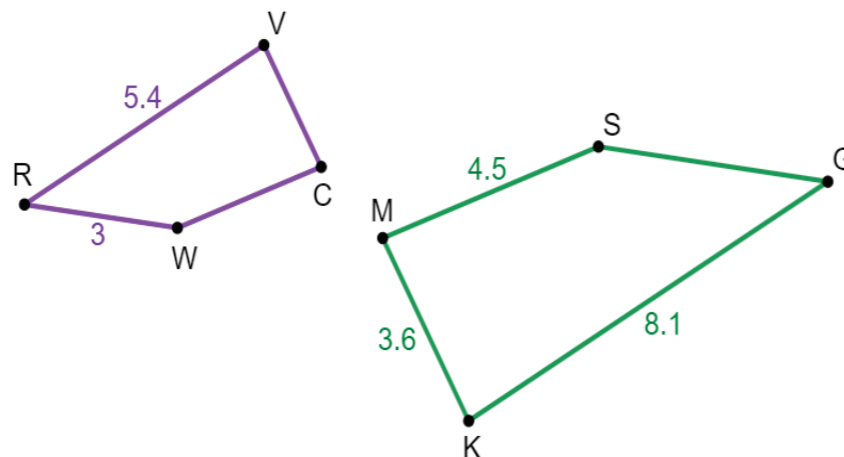
- c. What is the ratio of the corresponding sides of $RFDE$ to $HTPY$?

- d. Determine each measure.

$$DF = \underline{\hspace{2cm}} \quad RF = \underline{\hspace{2cm}}$$

$$YP = \underline{\hspace{2cm}}$$

3. Kirstyn and Roberto were finding the missing side measures for $RVCW$ and $GKMS$. They were given $RVCW \sim GKMS$, $VR = 5.4$, $RW = 3$, $SM = 4.5$, $MK = 3.6$, and $KG = 8.1$.



Kirstyn's and Roberto's work to find the measure of $\overline{VC}$ is shown.

Kirstyn's Work	Roberto's Work
$\frac{8.1}{5.4} = 1.5$	$\frac{8.1}{5.4} = 1.5$
$3.6 \times 1.5 = 5.4$	$3.6 \div 1.5 = 2.4$
$VC = 5.4$	$VC = 2.4$

- a. Who do you agree with?
- b. Explain the error in the incorrect work.
- c. Complete the statements.

$$CW = \underline{\hspace{2cm}} \quad SG = \underline{\hspace{2cm}}$$



6.10.4 Wrap-Up: Reflection

1. One thing I struggled with today was ...
2. One thing I need to practice more is ...
3. One thing I understand very well is ...



Unit 6, Lesson 11: Solving Problems Using Similarity – Part 1



Benchmark

MA.912.GR.1.6 Solve mathematical and real-world problems involving congruence or similarity in two-dimensional figures.



Learning Target

- ☐ I can solve mathematical problems involving similarity in two-dimensional figures.